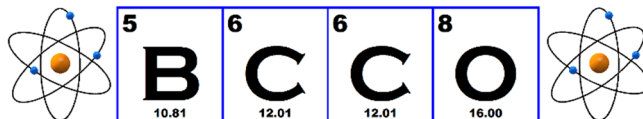


JUNIOR CANADIAN CHEMISTRY OLYMPIAD

British Columbia Chemistry Olympiad



2021

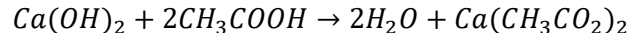
The Periodic Table/Data Sheet may be used for all parts of the contest, and non-programmable calculators are allowed.

Write your answers on the back side of the Answer Sheet provided. Do not forget to write your name and your school name on the front of the Answer Sheet. While students are expected to attempt all questions for a complete examination in 1.5 hours, it is recognized that backgrounds will vary and students will not be eliminated from further competitions because they have missed parts of the paper.

- 1) The highest-grade uranium deposits in the world are found in the Athabasca Basin in Canada with an average of 18% by weight uranium. If a mine in the Athabasca Basin collects 1 metric ton (1000 kg) of ore, how many moles of uranium can be extracted from the ore?

a) 4200 b) 420 **c) 756** d) 7560 e) 4.20

- 2) Several general types of chemical reactions can occur based on what happens when going from reactants to products. What is the type of the following reaction?



a) Synthesis
b) Single replacement
c) Decomposition
d) Neutralization
e) Combustion

- 3) Calculate the mass of potassium iodide (KI) required to prepare a 0.30 M solution in water using a 250 mL volumetric flask?

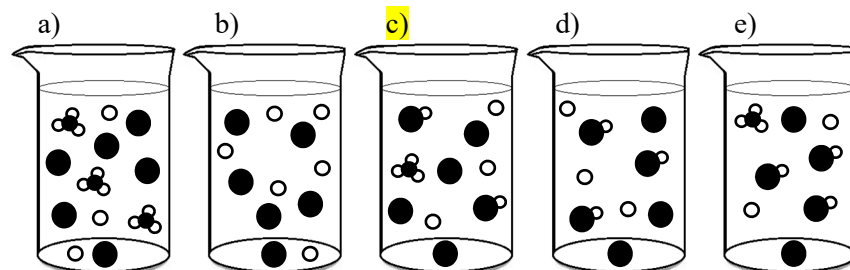
a) 5.98 g b) 199.2 g **c) 12.45 g** d) 49.80 g e) 24.90 g

- 4) Which of the following chemical equations is properly balanced?

a) $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$
b) $\text{C}_6\text{H}_8 + 6\text{O}_2 \rightarrow 5\text{CO}_2 + 4\text{H}_2\text{O}$
c) $\text{CH}_3\text{CH}_2\text{OH} + 3\text{O}_2 \rightarrow \text{CO}_2 + 3\text{H}_2\text{O}$
d) $\text{CH}_3\text{OH} + \text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
e) $2\text{C}_7\text{H}_5\text{N}_3\text{O}_6 \rightarrow 3\text{N}_2 + 5\text{H}_2 + 12\text{CO} + 2\text{C}$

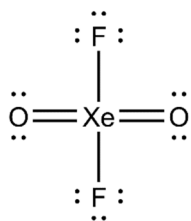
- 5) A 250 mL beaker is filled with 100 mL of a 0.30 M solution of acetic acid (CH_3COOH) and 150 mL of a 0.10 M solution of sodium hydroxide. Acetic acid is a weak acid and all solutions have been prepared in water. Which beaker diagram is the best representation of the solution obtained after the reaction is complete?

Legend: Na^+ ○ CH_3COO^- ●
 H_3O^+ ○● CH_3COOH ○●



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- 6) The density of chloroform (CHCl_3) at 25°C is 1.4788 g/cm^3 . What is the volume of 1 mole of chloroform at 25°C ?
 a) 34.1 mL **b) 80.7 mL** c) 176.5 mL d) 17.7 mL e) 56.7 mL
- 7) The province of Saskatchewan is one of the world's biggest producer of potash (a source of potassium used as soil fertilizer). Potash amounts are reported in K_2O equivalent, although it never contains potassium oxide. The K_2O equivalent system is used to compare weights of fertilizers because it is only the potassium weight that matters, the weight of the anion is irrelevant. If 200 tons of K_2CO_3 were extracted from a mine, what would be the K_2O equivalent weight in tons?
 a) 113.2 b) 293.4 **c) 136.3** d) 353.5 e) 56.6
- 8) Octane rating is used as a standard measure of the performance of a car engine. The octane isomer 2,2,4-trimethylpentane (C_8H_{18}) is the 100% reference standard for octane rating and has a density of 0.690 g/cm^3 . If 1 litre of 2,2,4-trimethylpentane is burned in excess of oxygen (O_2) how many grams of CO_2 is produced?
 a) 352.1 b) 913.8 c) 5520 **d) 2127** e) 1791
- 9) The Lewis structure of XeO_2F_2 shown below is not the best. Explain by selecting the best answer from the list.



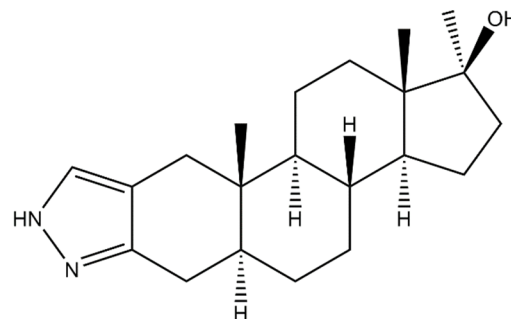
- a) The total number of electrons is incorrect**
 b) The central atom cannot be hypervalent
 c) A terminal atom is hypervalent
 d) The formal charges are calculated correctly but not minimized
 e) Noble gas never make compounds

- 10) Effective nuclear charge (Z_{eff}) is a useful tool for understanding periodicity. Z_{eff} is the net positive charge experienced by an electron in a polyelectronic atom. The term "effective" is used because the positive charge of the nuclei cannot reach a given electron completely, due to the shielding effect of other electrons. Z_{eff} is a theoretically calculated value (not an atomic property) but a good estimate of Z_{eff} can be obtained using Slater's rule. Slater's rule states that $Z_{\text{eff}} = Z - S$ where Z is the atomic number of the atom and S a screening constant. In the second row of the periodic table, the screening value is 0.85 for a core electron and 0.35 for a valence electron. The screening constant (S) is obtained by the summation of all the screening values of other electrons (excluding the electron for which the effective nuclear charge is to be determined). Use Slater's rule to estimate the effective nuclear charge for a valence electron in fluorine (F).
 a) 4.85 **b) 5.20** c) 4.15 d) 3.80 e) 8.65
- 11) To study weather phenomena one uses sensors carried by balloons. On a beautiful summer day when the temperature on the ground is 25°C and the pressure is 101.325 kPa, a balloon is partially inflated with 70 m^3 of helium. The balloon is released and climbs to an altitude of 10 kilometers. Calculate the volume of the balloon if the temperature has dropped to -50.0°C and the pressure to 27.5 kPa. Assume ideal gas behavior.
a) 193.0 m^3 b) 93.4 m^3 c) 52.4 m^3 d) 25.4 m^3 e) 257.9 m^3
- 12) What is the molecular geometry of NH_3 ?
 a) Trigonal Planar
 b) Trigonal Bipyramidal
 c) Square Pyramidal
d) Trigonal pyramidal
 e) Tetrahedral

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- 13) Which of the following element is expected to be the most electronegative?
a) Lu **b) Ts** c) Ra d) Lr e) U
- 14) Electroplating has many useful applications. It can improve a metal's surface appearance, electrical conductivity or anti-corrosion properties. 49.2 g of a copper salt is dissolved in 250 mL of water and the solution is used for electroplating until all the copper is reduced. After the experiment it is found that the cathode weight as increased by exactly 14.0 g. What is the chemical formula of the copper salt used?
a) CuCl_2 b) CuSO_4 c) CuSO_3 **d) CuBr_2** e) $\text{Cu}(\text{NO}_3)_2$
- 15) Which of the following molecules can form a hydrogen bond?
a) HBr **b) HF** c) CH_4 d) H_2CO e) CF_3H
- 16) After the earthquake and tsunami disaster, the soil, the air, and the water near the Fukushima Daiichi Nuclear Power Plant in Japan were found to be contaminated with radioactive ^{137}Cs , which has a half-life of 30.1 years. Radioactive decay is a first order process. How many years must pass before the radioactivity drops to 12.5% of its initial value?
a) 240.8 b) 376.3 c) 60.2 d) 120.4 **e) 90.3**
- 17) Sodium benzoate is produced by the neutralization of benzoic acid and is commonly used as a pH adjustor and preservative in food. Benzoic acid has a pK_a of 4.2 and is an aromatic carboxylic acid. What is the pH of a 0.10 M solution of benzoic acid?
a) 2.6 b) 3.1 c) 3.6 d) 5.2 e) 1.0

- 18) Most serious athletes will tell you that the drive to win is fierce. Besides the satisfaction of personal accomplishment, Olympic athletes often pursue dreams of winning a medal for their country. In such an environment, the use of anabolic steroids has become increasingly common. Anabolic steroids are synthetic variations of the male hormone testosterone. Which of the following molecular formula is the correct one for the steroid shown below?



- a) $\text{C}_{20}\text{H}_{26}\text{N}_2\text{O}$
b) $\text{C}_{18}\text{H}_{32}\text{N}_2\text{O}$
c) $\text{C}_{21}\text{H}_{28}\text{N}_2\text{O}$
d) $\text{C}_{21}\text{H}_{32}\text{N}_2\text{O}$
e) $\text{C}_{18}\text{H}_{26}\text{N}_2\text{O}$
- 19) An unknown liquid organic compound contains carbon, hydrogen and oxygen only. It has a very high normal boiling point of 205.8°C . 1.00 gram of this compound undergoes complete combustion reaction to produce 2.27g of carbon dioxide and 0.931g of water. When this compound is vaporized at its normal boiling point, the gaseous phase has a density of 2.96 g/L. Based on the information given, this compound is
a) Acetone
b) Propanal
c) Cyclopropanol
d) Hexanoic acid
e) Ethyl Butanoate
- 20) Which of the following the following atoms or ions has the largest radius?
a) Fr^+ b) Br^- c) Og d) Kr **e) At^-**

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- 21) How much energy is needed to convert 50.0 grams of ice from -10°C to a steam of 110°C . The specific heat is 2.108J/g for ice, 4.184J/g for water and 1.996J/g for steam. The heat of fusion is 6.02kJ/mol for ice is and heat of vaporization is 40.7kJ/mol for water.
- a) 152.6kJ b) 1526J c) 2359J d) 235.9kJ e) 23.59kJ
- 22) When the intermolecular force of a liquid increases, which of the following statement is incorrect?
- a) Boiling point increases
 b) Viscosity increases
 c) Surface tension increases
 d) Vapour pressure increase
 e) Density increases
- 23) Which of the following listed particles are isoelectronic?
- a) C, N, O, F, Ne
 b) B, Si, As, Te, At
 c) Mg^{2+} , Na^{+} , Ne, F^{-} , O^{2-}
 d) C, Si, Ge, Sn, Pb
 e) Mn, Fe, Co, Ni, Cu
- 24) Diethyl ether and butanol are two organic isomers. What do they have in common under the same temperature and pressure?
- a) Solubility in water
 b) Heat of vaporization
 c) Molar volume
 d) Reactivity
 e) Gaseous state density
- 25) A hydrocarbon mixture known to contain methane, ethene and butane only has a volume of 6.00L , a pressure of 2 bar and density of $2.404 \times 10^{-3}\text{g/mL}$ at 25°C . When this mixture reacted with hydrogen gas in the presence of platinum metal, 0.3145g of hydrogen gas was consumed. When the same amount of the mixture was completely combusted, 43.13g of carbon dioxide and 23.69g of water were produced. What was the mass percentage of methane in the mixture? (assume all reactions went to completion).
- a) 19 % b) 24 % c) 30 % d) 45 % e) 50 %

End of examination questions.

